

60–70% of EV batteries get destroyed.

Not because they're bad.

Because nobody trusts the data.

€1–3 Billion destroyed annually. We fix the trust layer.

The Battery Data Crisis

- EU Reg. 2023/1542 mandates a Digital Product Passport for every EV battery
- 102 data points: carbon footprint, health, recycled content, supply chain
- **Manufacturers self-report. Nobody cross-validates.**
- Buyers can't trust SoH claims. Regulators can't enforce compliance.
- **Result: batteries are scrapped instead of reused — billions wasted.**

Three Stakeholders. One Missing Layer.



Business

- Cannot verify supplier data
- No ERP action on bad actors
- Compliance risk from 2027
- Cross-border data inconsistency



Regulator

- Cannot detect who falsifies data
- No centralized violation tracking
- Manual audits only
- No real-time EU-wide view



Consumer

- Pays €10K+ with no reliable health info
- No trusted certificate at point of sale
- Cannot assess second-life value
- No recourse if data was wrong

Existing Tools Store Data. Nobody Validates It.

Tool	What It Does	What's Missing
Minespider	Blockchain supply chain storage	No cross-field validation
Spherity	Digital identity credentials	No trust scoring or ERP action
Circular	Traceability platform	No procurement decision output
Raw DPP	Self-reported data container	Anyone can type any number

The data exists. The trust doesn't.

TrustLayer VeriCell

AI-powered trust validation for battery DPP data —
turning unverified passports into live ERP decisions.



12
34

Trust Score

0 – 100



Decision

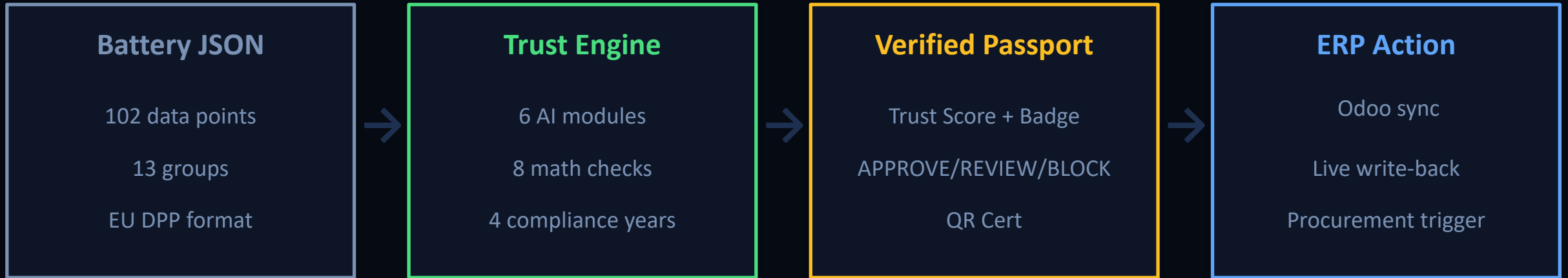
APPROVE / REVIEW / BLOCK



ERP Action

Live sync to procurement

The Validation Pipeline



6 Engine Modules:

- Sustainability Scorer (5 sub-scores, weighted)
- Authenticity Checker (8 cross-field math checks)
- Compliance Tracker (2024 / 2027 / 2031 / 2036)
- Sensor Validator (voltage, resistance, temp, SoC)
- Trust Score Calculator (data quality + evidence)
- SoH Composite (5 health parameters)

Full EU 2023/1542 Coverage

13 Data Groups (A–M)

- A — General Info
- B — Carbon Footprint
- C — Recycled Content
- D — Responsible Sourcing
- E — Electrical
- F — Conformity & Labelling
- G — Composition
- H — Authority Information
- I — Individual Perf.
- J — State of Health
- K — Expected Lifetime
- L — Operational Data
- M — Identification

Compliance Timeline

2024	✓	DoC, CE marking, labelling
2027	⚠	QR code, carbon declaration (Art. 7)
2031		Recycled Co ≥16%, Li ≥6%, Ni ≥6% (Art. 8)
2036		Co ≥26%, Li ≥12%, Ni ≥15%

102 data points · 4 compliance horizons · 3 chemistry types · 13 data groups

What Makes It Work



Trust Score That Can't Be Gamed

Math-enforced ceiling: $\text{max trust} = (\text{real_data\%} \times 0.5) + 50$
A supplier with 0% verified data cannot score above 50/100.



8 Cross-Field Authenticity Checks

$V \times Ah = kWh$ | Carbon class must match declared value
Cell count must add up | Supply chain verifier must be named.



Role-Based Access — 4 Levels

Consumer: health + QR certificate
Technician: sensor data + disassembly
Company: supply chain risk + compliance
Regulator: all 102 points, unredacted.



Live ERP Sync (Odoo)

BLOCK decision → writes to ERP product record in real-time
No manual entry. Procurement trigger is automatic.

Live Demo — 3 Batteries, Real Decisions

 **1 SUPPLIER BLOCKED** — Immediate procurement action required

Supplier	Trust Score	Decision	2024 Status	Auth Checks
Volvo Cars (NMC811)	71.5 / 100	 APPROVE	COMPLIANT	8/8 
BMW Group (NMC712)	65.5 / 100	 REVIEW	COMPLIANT	6/8 
Shenzhen Generic (LFP)	6.0 / 100	 BLOCK	AT RISK	2/5 

→ Drill down → Trust Score gauge → Sync to Odoo ERP → toast confirmation

→ Government Portal → EU compliance map → violation chart by EU Article

→ Battery Passport → Consumer → QR Certificate (Level 2 Physical Verified badge)

[Replace this area with a live screenshot of the Enterprise Dashboard]

Built to Be Real

Stack

- Python 3.11 + Streamlit 1.32
- Plotly 5.18 (charts + EU map)
- xmlrpc.client — stdlib only
- qrcode[pil] (QR certificates)
- python-pptx (this deck!)

Data Layer

- 3 manufacturer JSON files
- EU DPP format (102 data points)
- Chemistry benchmarks: NMC811, NMC712, LFP
- EU compliance thresholds per year
- Live Odoo: vericell.odoo.com

Smart Logic

- @st.cache_data — engine runs once
- Deterministic + fully auditable
- ERP adapter pattern (swap Odoo→SAP = config change)
- Trust ceiling formula — ungameable
- Cross-field math — 8 validation checks

Our Unfair Advantage

Only platform that outputs ERP actions

Not dashboards — actual procurement triggers. BLOCK → Odoo writes in real-time.

ERP-agnostic architecture

Same JSON output works with Odoo, SAP, weclapp, Dynamics. Adapter swap = config change.

Math-enforced trust scoring

Score formula cannot be gamed. 0% verified data = max score of 50/100. Period.

Private trust layer — no regulatory approval needed

Like AVILOO for used EVs. We issue the badge. Buyers pay more for certified batteries.

Right timing — mandatory from 2027

EU DPP is law. Every battery maker needs validation. We build the layer now.

IMPACT- What Changes



For Companies

- Automated supplier risk scoring
- ERP blocks bad batteries before procurement
- Compliance evidence ready for audits
- Real-time decision — no manual review



For Regulators

- Real-time violation tracking dashboard
- EU compliance map by country
- Downloadable audit reports per battery
- Cross-field fraud detection — automated



For the Market

- More batteries reused, not destroyed
- €3B+ inefficiency reduced
- Trusted second-life battery economy
- EU 2027 compliance — ready now

Where This Goes

NOW

Phase 1

- Trust Engine + Odoo ERP
- 3 manufacturer demo
- 4-role access system
- EU 2023/1542 compliance

6 months

Phase 2

- API upload portal for manufacturers
- SAP / weclapp integrations
- Level 3: TÜV co-signed certs
- Onboard first paying customer

12 months

Phase 3

- Insurance & leasing data feed
- White-label engine for ERP vendors
- Solar + industrial battery expansion
- Pan-EU rollout

Revenue Model

€30–80 / battery

per validation (second-life operators)

€1.5K–5K / month

SaaS license (fleet operators, recyclers)

€50K–500K / year

enterprise + insurance data contracts

Manufacturers claim
Regulators demand proof.
Buyers need certainty.

We provide it — from algorithm to physical sensor.

The problem costs €3 billion a year.

**TrustLayer VeriCell — Trust the battery. Not just
the paperwork.**